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Project Title	Week 3: Churn Prediction for StreamWorks Media

1. Introduction

StreamWorks Media is a UK-based subscription video streaming platform operating in a highly competitive market against global players such as Netflix and Amazon Prime, and as customer acquisition becomes more expensive, the company is shifting its focus to retaining existing users and reducing churn; currently, it faces an overall churn rate of about 23% across 1,500 subscribers, who are mostly middle-aged (average age ~44), pay around £10 per month, and watch on average 40 hours of content each month.

2. Data & Methodology

Dataset Overview

The dataset `streamworks_user_data.csv` contains 1,500 subscribers and 14 columns.

Key descriptive statistics:

- Age: mean 43.7 years (range 18–69)
- Average watch hours / month: mean 39.9 (IQR ~19–60 hours)
- Mobile app usage: mean 51.4% of total viewing
- Complaints: mean 2.5 complaints per user
- Monthly fee: mean £10.18 (tiers around £5.99, £9.99, £13.99, £14.99)
- Churn: mean 0.234 → 23.4% churn rate

Data Cleaning & Feature Engineering

- Converted `signup_date` and `last_active_date` to datetime and created `tenure_days` and `is_loyal` (`tenure > 180` days).
- Imputed missing numeric values with medians and categorical features with modes; removed records with missing `user_id` and `is_churned`.
- Engineered key features:
 - `watch_per_fee_ratio` (value for money),
 - `heavy_mobile_user` (mobile usage $\geq 75\%$),
 - `low_watch_time` (below median watch),
 - `promo_and_low_watch` (received promotion AND low watch time),
 - Binned age into age groups.
- Applied one-hot encoding to gender, country, `subscription_type` and binary mappings for promotions and referrals.

Modelling Approach

For churn prediction (classification):

- Split into train (80%) / test (20%), stratified by churn.
- Standardised numeric features; tested:
 - Logistic Regression with class weights
 - Random Forest with class weights
 - SVM (RBF) with class weights

For engagement prediction (regression):

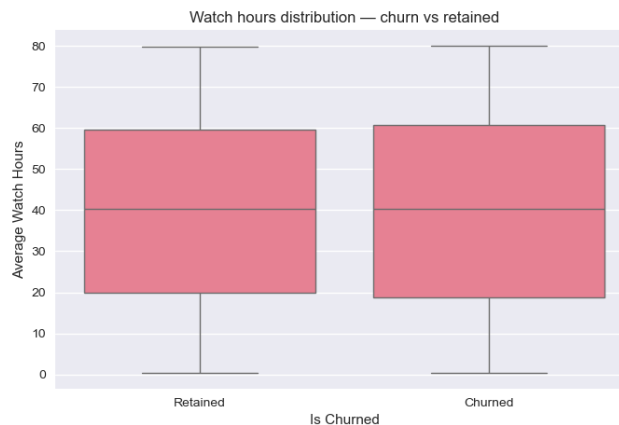
- Linear regression to predict average_watch_hours from transformed numeric and encoded categorical features.

3. Exploratory Data Analysis (EDA)

Customer Profile

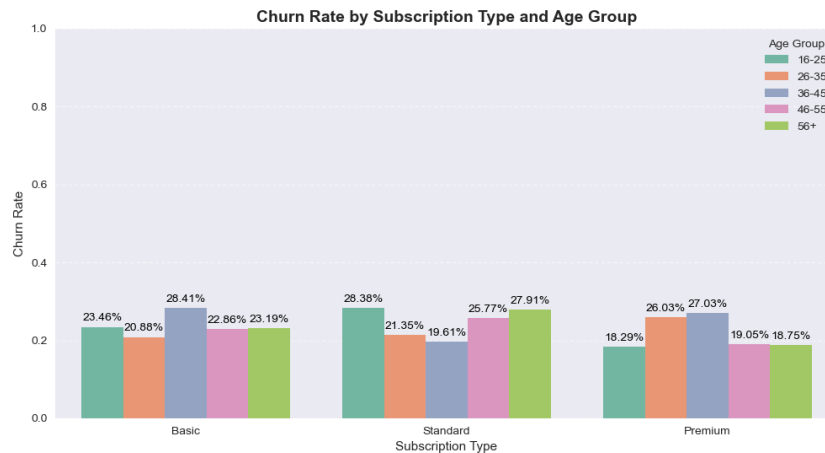
StreamWorks serves a diverse audience: multiple countries (with Canada as the largest single market), all three subscription tiers (with Basic the most common), and a mix of genders.

Average usage (~40 hours per month) suggests fairly engaged subscribers, though there is a wide spread from very light to heavy viewers.



Churn Patterns by Segment

Segmentation by subscription type and age group reveals that churn rates range roughly from ~21% to ~28% across major segments ($n \geq 30$). Some Standard segments (e.g. ages 16–25 and 56+) show churn at around 27–28%, slightly above the platform average, while many Basic segments sit closer to 23–24% and Standard 26–35 around 21%.



4. Statistical Analysis

Chi-Square Tests (Categorical Drivers of Churn)

Chi-square tests were applied to `received_promotions`, `referred_by_friend`, and `gender` dummies:

- Received promotions: $p \approx 0.115 \rightarrow$ no strong evidence that churn is statistically different between customers who received promotions and those who did not, although observed churn is somewhat lower among promotion recipients (21.5% vs 25.1%).
- Referrals: $p \approx 0.439 \rightarrow$ no significant difference in churn between referred and non-referred users.
- Gender (Female indicator): $p \approx 0.041 \rightarrow$ a statistically significant, but small, association between being female and churn.

T-Test (Watch Time vs Churn)

A t-test comparing average watch hours for churned vs retained users shows:

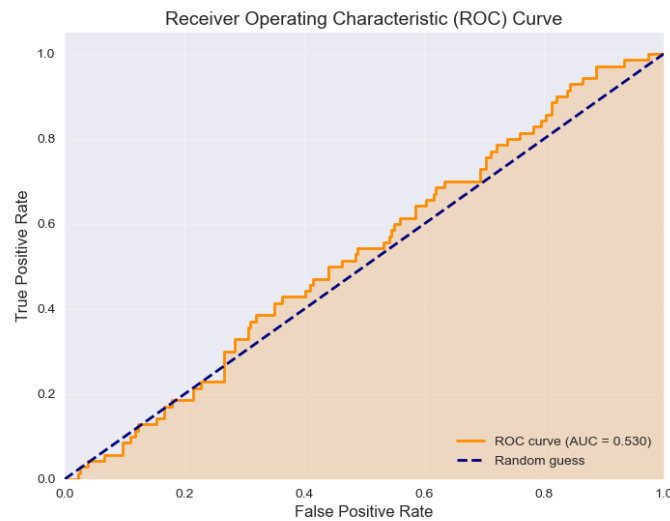
- Mean watch hours: 39.8 (churned) vs 40.0 (retained)
- $t \approx -0.15$, $p \approx 0.88 \rightarrow$ difference is not statistically significant.

5. Predictive Modelling

Churn Classification Models

- Logistic Regression: Accuracy ~ 0.68 ; ROC-AUC ≈ 0.53 ; recall for churners remains low despite class weighting.

- Random Forest: Slightly better with ROC-AUC ≈ 0.55 , but still only marginally above random. Confusion matrix reveals many churners are still misclassified as non-churners.
- SVM (RBF): ROC-AUC ≈ 0.49 , essentially no better than random.

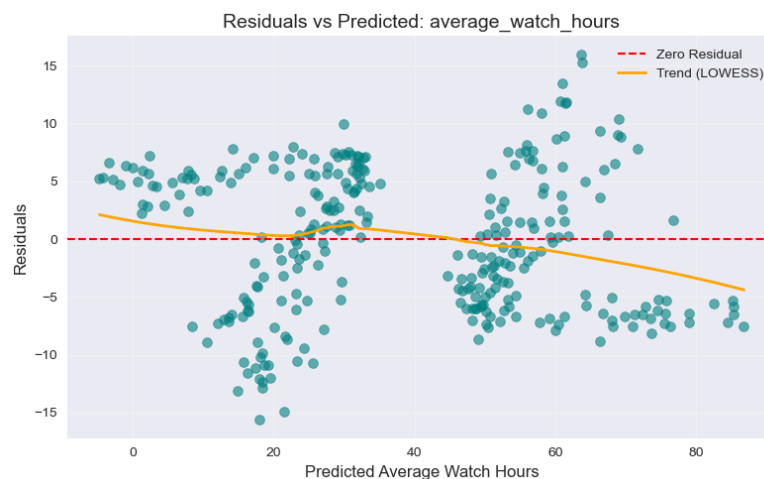


Watch-Time Regression Model

A linear regression model for average_watch_hours performs well:

- RMSE ≈ 6.20 hours, $R^2 \approx 0.92$, indicating that most of the variability in watch hours can be explained by the features.

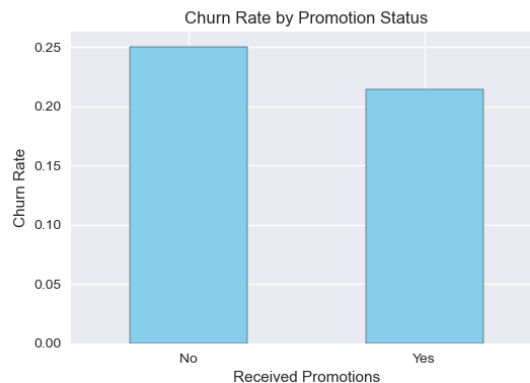
Top predictors of watch time include watch_per_fee_ratio, monthly_fee, subscription type, and tenure_days.



6. Business Question Answers

1. Do users who receive promotions churn less?

The problem is to know if promotions genuinely reduce churn or just discount revenue. This matters because promotions cost money and should only be used when they change behaviour. Our analysis shows churn is slightly lower for promotion recipients (21.5% vs 25.1%), but the effect is not statistically significant ($p \approx 0.115$), so evidence is weak. To solve this, StreamWorks should stop broad, generic offers and instead target promotions at at-risk segments (e.g. short-tenure, low-watch users) and A/B test tailored offers to confirm real churn reduction before scaling.

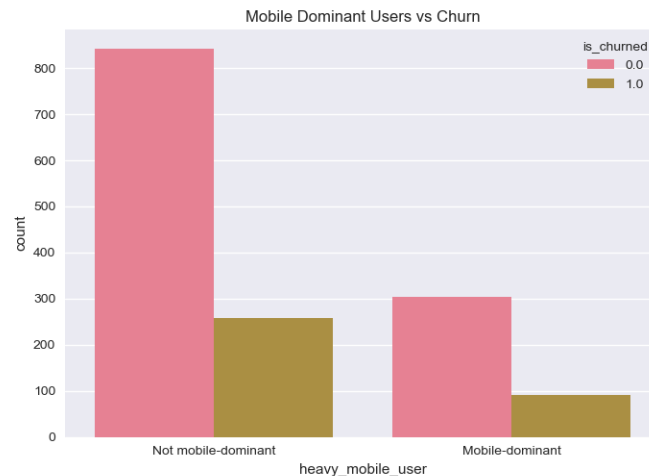


2. Does watch time impact churn likelihood?

The problem is to know whether low watch time is a reliable early warning sign for churn. This matters because if watch time strongly predicted churn, StreamWorks could trigger simple, usage-based interventions when activity drops. Our t-test shows almost no difference in average watch hours between churned and retained users (≈ 39.8 vs 40.0 , $p \approx 0.88$), so watch time alone is not a clean churn trigger. To solve this, watch time should be combined with tenure and plan type (e.g. "low watch in first 90 days on Standard") and compared to a user's normal pattern to define richer risk scores and drive targeted nudges.

3. Are mobile-dominant users more likely to cancel?

The problem is whether heavy mobile usage signals higher churn risk or just a different consumption channel. This matters because it affects where StreamWorks invests (mobile UX vs TV/web) and how it designs notifications and in-app journeys. The chi-square test shows almost identical churn rates for mobile-dominant and non-mobile users ($p \approx 0.88$), so mobile use alone is not a strong risk factor. To solve this, StreamWorks should treat mobile dominance as a preference, improve the mobile experience generally, and use mobile behaviour as one signal within a broader churn model rather than as a standalone red flag.



4. What are the top features influencing churn?

The problem is identifying which factors truly drive churn so interventions focus on the right levers. This matters because acting on the wrong drivers wastes budget and effort. The random forest model shows churn is most strongly associated with `mobile_app_usage_pct`, `watch_per_fee_ratio`, and `average_watch_hours`, with `tenure` and `subscription_type` as secondary influences. To solve this, StreamWorks should monitor these variables closely and run experiments that improve perceived value (`watch_per_fee_ratio`), optimise mobile experience, and support low-engagement users before they cancel.

5. Which customer segments should be prioritised for retention?

The problem is deciding which groups should receive the most retention spend. This matters because losing some segments (e.g. older, long-tenure users) is more costly than others. Segmenting by subscription type and age shows churn is highest in Basic 36–45 and Standard 16–25 and 56+, while other Basic and Standard groups sit nearer the average. To solve this, StreamWorks should prioritise these high-churn segments with targeted engagement (onboarding flows, in-app guidance, surprise-and-delight offers) and design bespoke retention journeys for high-value Premium users.

6. What factors drive viewing time and tenure (engagement)?

The problem is understanding what drives higher viewing time and longer tenure, since engagement is closely linked to churn and lifetime value. This matters for decisions on content, pricing, and product features. The watch-time regression ($R^2 \approx 0.92$) shows that value-for-money (`watch_per_fee_ratio`), monthly fee tier, subscription type, and `tenure_days` strongly influence viewing hours. To solve this, StreamWorks should clearly communicate plan value, push content that sustains long-term viewing per segment, and run experiments (bundles, recommendations, loyalty rewards) that increase `watch_per_fee_ratio` and thereby strengthen perceived value and retention.